

1. Install nextflow and docker

Last login: Tue Apr 14 16:14:03 on ttys005

```
[(base) dariaandreeva@MacBook-Air-Daria-2 ~ % nextflow -version
```

```
N E X T F L O W
version 25.10.4 build 0
created 10-02-2026 14:37 UTC (17:37 MSK)
cite doi:10.1038/nbt.3820
http://nextflow.io
```

```
(base) dariaandreeva@MacBook-Air-Daria-2 ~ %
```

```
(base) dariaandreeva@MacBook-Air-Daria-2 ~ % brew install --cask docker
✓ JSON API cask.jws.json                               Downloaded 15.4MB/ 15.4MB
✓ JSON API formula.jws.json                             Downloaded 32.0MB/ 32.0MB
==> Fetching downloads for: docker-desktop
✓ API Source docker-desktop.rb                           Verified 5.1KB/ 5.1KB
✓ Cask docker-desktop (4.69.0,224084)                     Verified 607.3MB/607.3MB
==> Installing Cask docker-desktop
==> Moving App 'Docker.app' to '/Applications/Docker.app'
==> Linking Bash Completion 'docker-compose.bash-completion' to '/opt/homebrew/e
==> Linking Bash Completion 'docker.bash-completion' to '/opt/homebrew/etc/bash_
==> Linking Fish Completion 'docker-compose.fish-completion' to '/opt/homebrew/s
==> Linking Fish Completion 'docker.fish-completion' to '/opt/homebrew/share/fis
==> Linking Zsh Completion 'docker-compose.zsh-completion' to '/opt/homebrew/sha
==> Linking Zsh Completion 'docker.zsh-completion' to '/opt/homebrew/share/zsh/s
==> Linking Binary 'docker-compose' to '/usr/local/cli-plugins/docker-compose'
Password:
==> Linking Binary 'hub-tool' to '/usr/local/bin/hub-tool'
==> Linking Binary 'kubect1' to '/usr/local/bin/kubect1.docker'
==> Linking Binary 'docker' to '/usr/local/bin/docker'
==> Linking Binary 'docker-credential-desktop' to '/usr/local/bin/docker-credent
==> Linking Binary 'docker-credential-ecr-login' to '/usr/local/bin/docker-crede
==> Linking Binary 'docker-credential-osxkeychain' to '/usr/local/bin/docker-cre
🔧 docker-desktop was successfully installed!
(base) dariaandreeva@MacBook-Air-Daria-2 ~ %
```

2. Check out rnaseq-nf pipeline

```
(base) dariaandreeva@MacBook-Air-Daria-2 ~ % git clone https://github.com/nextflow-io/rnaseq-nf.git
Cloning into 'rnaseq-nf'...
remote: Enumerating objects: 1036, done.
remote: Counting objects: 100% (636/636), done.
remote: Compressing objects: 100% (241/241), done.
remote: Total 1036 (delta 416), reused 536 (delta 343), pack-reused 400 (from 2
Receiving objects: 100% (1036/1036), 665.41 KiB | 2.20 MiB/s, done.
Resolving deltas: 100% (507/507), done.
(base) dariaandreeva@MacBook-Air-Daria-2 ~ %
```

3. Run the pipeline on the test data in conda and Docker

```
(base) dariaandreeva@MacBook-Air-Daria-2 ~ % nextflow run nextflow-io/rnaseq-nf --profile conda --resume
N E X T F L O W ~ version 25.10.4
Launching 'https://github.com/nextflow-io/rnaseq-nf' [dreamy_pasteur] DSL2 - revision: ca20a6dfd2 [master]
R N A S E Q ~ N F P I P E L I N E
=====
transcriptome: /Users/dariaandreeva/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal171.500bpflank.fa
reads         : /Users/dariaandreeva/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_(1,2).fq
outdir        : results
[~          ] RNASEQ:INDEX -
[~          ] RNASEQ:FASTQC -
executor > local (3)
executor > local (4)
[0d/0707ba] RNA_(ggal_1_48850000_49020000) | 1 of 1 ✓
[b9/e444ce] RNASEQ:FASTQC [ggal_gut]       | 1 of 1 ✓
[0d/1dc2b1] RNASEQ:QUANT [ggal_gut]        | 1 of 1 ✓
[e8/964d95] MULTIQC                       | 1 of 1 ✓
Done! Open the following report in your browser -> results/multiqc_report.html
Completed at: 14-Apr-2026 18:04:56
Duration    : 16m 26s
CPU hours   : (a few seconds)
Succeeded   : 4
(base) dariaandreeva@MacBook-Air-Daria-2 ~ %
```

```

nextflow run nextflow-io/rnaseq-nf --profile docker --c docker_amd64.config --resume

N E X T F L O W  ~ version 25.10.4

Launching 'https://github.com/nextflow-io/rnaseq-nf' [special_mahavira] DSL2 ~ revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
=====
transcriptome: /Users/dariaandreeva/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads         : /Users/dariaandreeva/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir        : results

executor > local (4)
[c2/e50f6a] RNA: (ggal_1_48850000_49020000) | 1 of 1 ✓
[e0/991d75] RNASEQ:FASTQC (ggal_gut)      | 1 of 1 ✓
[f1/7d3184] RNASEQ:QUANT (ggal_gut)       | 1 of 1 ✓
[ae/593f95] MULTIQC                       | 1 of 1 ✓

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 14-Apr-2026 22:46:16
Duration     : 1m 53s
CPU hours    : 0.1
Succeeded    : 4

(base) dariaandreeva@MacBook-Air-Daria-2 rnaseq_nf_runs % █

```

4. Run the pipeline on my data

I used SRR384939_{1,2}.fastq (Drosophila melanogaster)

```

--outdir results_SRR384939_dmel \
--work-dir work_SRR384939_dmel

N E X T F L O W  ~ version 25.10.4

Launching 'https://github.com/nextflow-io/rnaseq-nf' [pensive_allen] DSL2 ~ revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
=====
transcriptome: /Users/dariaandreeva/Desktop/Итмощная сванка/pipline/hwi/rnaseq_nf_runs/references/dmel_cdna.fa
reads         : ncbi_data/SRR384939_{1,2}.fastq
outdir        : results_SRR384939_dmel

executor > local (3)
[5c/8c5d2c] RNASEQ:INDEX (dmel_cdna)      | 1 of 1 ✓
[0d/3ceb03] RNASEQ:FASTQC (SRR384939)     | 1 of 1, cached: 1 ✓
[7f/291c08] RNASEQ:QUANT (SRR384939)     | 1 of 1 ✓
[71/87ec03] MULTIQC                      | 1 of 1 ✓

Done! Open the following report in your browser --> results_SRR384939_dmel/multiqc_report.html

Completed at: 14-Apr-2026 22:09:39
Duration     : 2m 11s
CPU hours    : 0.1 (32,1% cached)
Succeeded    : 3
Cached       : 1

(base) dariaandreeva@MacBook-Air-Daria-2 rnaseq_nf_runs % █

```

5. Create new Github repo

Done

6. Upload

Done